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Title: Management of Community - Acquired Pneumonia in Infants and Children

Statement:

This policy helps pediatricians in acute care and sub-specialties in the management of children with community acquired pneumonia (CAP) as inpatient or outpatient sittings

Note: This guideline **does not** discuss the management of infants less than3 months of age or immunocompromised children.

Introduction:

Most common cause of pneumonia in preschool-aged children are viral that does not require antibiotics; but be aware of secondary bacterial infection following viral pneumonia. Therefore, X-rays or antibiotics are unnecessary unless child present with atypical presentation or other differential diagnosis is considered.

Community acquired pneumonia is defined as community acquired infection in a previously healthy child.

Admission criteria of a child with CAP:

- Children and infants who have moderate to severe CAP, as defined by several factors, including respiratory distress and hypoxemia (sustained saturation of peripheral oxygen [SpO2] <90 % at sea level). (Strong recommendation; high-quality evidence). refer to table 1
- Infants less than 3–6 months of age with suspected bacterial CAP are likely to benefit from hospitalization. (Strong recommendation; low-quality evidence)
- Children and infants with suspected or documented CAP caused by a pathogen with increased virulence, such as community-associated methicillin-resistant Staphylococcus aureus (CA-MRSA) should be hospitalized.
- Children and infants for whom there is concern about careful observation at home or who are unable to comply with therapy or unable to be followed up should be hospitalized. (Strong recommendation; low-quality evidence)

Table 1 Criteria for CAP Severity of Illness in Children with

Community-Acquired Pneumonia

Major criteria

- Invasive mechanical ventilation
- Fluid refractory shock
- Acute need for NIPPV
- Hypoxemia requiring FiO_2 greater than inspired concentration or flow feasible in general care area

Minor criteria

- Respiratory rate higher than WHO classification for age
- Apnea
- Increased work of breathing (eg, retractions, dyspnea, nasal flaring, grunting)
- $\text{PaO}_2/\text{FiO}_2$ ratio <250
- Multilobar infiltrates
- PEWS score >6
- Altered mental status
- Hypotension
- Presence of effusion
- Comorbid conditions (eg, HgbSS, immunosuppression, immunodeficiency)
- Unexplained metabolic acidosis

Modified from Infectious Diseases Society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults [27, table 4]. Clinician should consider care in an intensive care unit or a unit with continuous cardiorespiratory monitoring for the child having ≥ 1 major or ≥ 2 minor criteria.

Abbreviations: FiO_2 , fraction of inspired oxygen; HgbSS, Hemoglobin SS disease; NIPPV, noninvasive positive pressure ventilation; PaO_2 , arterial oxygen pressure; PEWS, Pediatric Early Warning Score [70].

Microbiologic Testing

Blood Cultures:

- 1- In the outpatient setting, blood cultures should not be routinely performed in nontoxic, fully immunized children with CAP. (strong recommendation; moderate-quality evidence).
- 2- Blood cultures should be obtained in children requiring hospitalization for presumed bacterial CAP that is moderate to severe, particularly those with complicated pneumonia. (Strong recommendation; low-quality evidence)
- 3- Blood cultures should be obtained in children who fail to demonstrate clinical improvement and in those who have progressive symptoms or clinical deterioration after initiation of antibiotic therapy (strong recommendation; moderate-quality evidence)

Sputum Gram Stain and Culture:

Sputum samples for culture and Gram stain should be obtained in hospitalized children who can produce sputum. (Weak recommendation; low-quality evidence)

Urinary Antigen Detection Tests

Urinary antigen detection tests are not recommended for the diagnosis of pneumococcal pneumonia in children; false-positive tests are common. (strong recommendation; high quality evidence)

Testing For Viral Pathogens

Sensitive and specific tests for the rapid diagnosis of influenza virus, SARS-CoV-2 and other respiratory viruses should be used in the evaluation of children with CAP. A positive influenza test may decrease both the need for additional diagnostic studies and antibiotic use, while guiding appropriate use of antiviral agents in both outpatient and inpatient settings. (Strong recommendation; high-quality evidence).

Note in our sitting it is recommended to send respiratory viral panel in patients admitted in high dependency or intensive care units)

Acute-Phase Reactants:

- 1- Acute-phase reactants, such as the erythrocyte sedimentation rate (ESR), C-reactive protein (CRP) cannot be used as the sole determinant to distinguish between viral and bacterial causes of CAP. (strong recommendation, high-quality evidence)

evidence)

Pulse Oximetry:

Pulse oximetry should be performed in all children with pneumonia and suspected hypoxemia. The presence of hypoxemia should guide decisions regarding site of care and further diagnostic testing. (Strong recommendation; moderate quality evidence)

Chest Radiographs:

1- Chest radiographs should be obtained in all patients hospitalized for management of CAP to document the presence, size, and character of parenchymal infiltrates and identify complications of pneumonia that may lead to interventions beyond antimicrobial agents and supportive medical therapy. (Strong recommendation; moderate-quality evidence)

2- Repeated chest radiographs should be obtained in children who fail to demonstrate clinical improvement and in those who have progressive symptoms or clinical deterioration within 48–72 hours after initiation of antibiotic therapy. (Strong recommendation; moderate-quality evidence)

3- Repeated chest radiographs 4–6 weeks after the diagnosis of CAP should be obtained in patients with recurrent pneumonia involving the same lobe and in patients with lobar collapse at initial chest radiography with suspicion of an anatomic anomaly, chest mass, or foreign body aspiration. (Strong recommendation; moderate quality evidence)

Antimicrobial therapy:

1- Inpatient sitting, amoxicillin should be used as first-line therapy for previously healthy, appropriately immunized infants and preschool and school age children with mild to moderate CAP suspected to be of bacterial origin. Amoxicillin provides appropriate coverage for *Streptococcus pneumoniae*, the most prominent invasive bacterial pathogen. (98% of pneumococcus in our isolates are covered by amoxicillin)

2- For inpatient sitting, please refer to table 2

3- Influenza antiviral therapy should be administered as soon as possible to children with moderate to severe CAP consistent with influenza virus infection during widespread local circulation of influenza viruses. Because early antiviral treatment has been shown to provide maximal benefit, treatment should not be delayed until confirmation of positive influenza test results. Treatment after 48 hours of symptomatic infection may still provide clinical benefit to those with more severe disease. (Strong recommendation; moderate-quality evidence)

4- Children on adequate therapy should demonstrate clinical and laboratory signs of improvement within 48–72 hours on appropriate antimicrobial therapy.

Table 2 antimicrobial choice in inpatient sitting

Age		Alternative
All ages if fully immunized including pneumococcal vaccine	Ampicillin or penicillin G (98.5% of streptococcus pneumonia are susceptible) do not use penicillin if meningitis is suspected ; Alternative: In complicated pneumonia or life-threatening infection ceftriaxone or cefotaxime; addition of vancomycin or clindamycin for suspected CA-MRSA	Azithromycin (in addition to b-lactam, if diagnosis of atypical pneumonia is in doubt); alternatives: clarithromycin or erythromycin; doxycycline for children > 7 years old; levofloxacin for children who have reached growth maturity, or who cannot tolerate macrolides
Not fully immunized or do not received pneumococcal vaccine	Cefuroxime or Augmentin Alternative for complicated pneumonia Cefotaxime/ ceftriaxone addition of vancomycin or clindamycin for suspected CA-MRSA	Same as above

Complicated pneumonia:

Complicated pneumonia commonly present as parapneumonic effusion but other form rarely encounter as lung abscess or necrotizing pneumonia. Antimicrobial therapy been mentioned in table 2.

For the Indications of drainage of parapneumonic effusion please see table3 below.

Table 3 Indications for drainage of parapneumonic effusion

		outcome	fibrinolysis
Small: <10mm on lateral decubitus radiograph or opacifies less than one-fourth of hemithorax	bacterial culture and gram stain results unknown or negative	low	No; sampling of pleural fluid is not routinely required
Moderate: >10mm of fluid but opacifies less than half of hemithorax	Bacterial culture and/or gram stain positive or negative(empyema)	Low to moderate	No; if the patient has no respiratory compromise and the pleural fluid is not consistent with empyema(sampling of pleural fluid by simple thoracentesis may help determine presence or absence of empyema and need for drainage procedure, and sampling with a drainage catheter may provide both diagnostic and therapeutic benefit; Yes; if patient has respiratory compromise or fluid is consistent with empyema
Large: opacifies more than half of hemithorax	Bacterial culture and/or gram stain positive(empyema)	high	Yes in most cases

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
1- The Management of Community-Acquired Pneumonia in Infants and Children Older Than 3 Months of Age: Clinical Practice Guidelines by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America.		2011	53(7):e25-e76
2- National public health laboratory IPD surveillance 2014-2016			
3- National antimicrobial guideline		2016-2017	
4- British thoracic Society guideline for the management of community acquired pneumonia in children		2011	

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